

MODULE 1: Overview of Internet & Application Layer

PCCST501 Computer Networks | Quick Exam Revision Guide | KTU S5 CSE | Prof. Anju Markose, VJCET

1. Overview of the Internet & Protocol Layering

Internet Components: The network is divided into the *Network Edge* (end systems, hosts, clients, servers) and the *Network Core* (mesh of interconnected routers and packet switches).

Packet Switching vs Circuit Switching:

- **Packet Switching:** Discrete data packets share network resources via *statistical multiplexing*. Highly efficient, handles bursty traffic, but introduces queuing delays and packet loss.
- **Circuit Switching:** Dedicated end-to-end communication channels reserved for the entire session (e.g. traditional PSTN phone network). Guaranteed bandwidth, but underutilized when idle.

Key Concept / Exam Formula:

Nodal Delay Formula: Total Delay = $d_{proc} + d_{queue} + d_{trans} + d_{prop}$

- **Transmission Delay (d_{trans}):** L / R (Packet length in bits / Link bandwidth in bps)
- **Propagation Delay (d_{prop}):** d / s (Physical distance in meters / Speed of light $\sim 2 \times 10^8$ m/s)

OSI vs TCP/IP Protocol Layering:

OSI Model (7 Layers)	TCP/IP Model (5 Layers)	Protocol Data Unit (PDU) & Examples
7. Application 6. Presentation 5. Session	5. Application Layer	Message: HTTP, FTP, SMTP, DNS
4. Transport Layer	4. Transport Layer	Segment / Datagram: TCP, UDP
3. Network Layer	3. Network Layer	Packet / Datagram: IPv4, IPv6, ICMP
2. Data Link Layer	2. Data Link Layer	Frame: Ethernet (802.3), WiFi (802.11)
1. Physical Layer	1. Physical Layer	Bits: UTP Cable, Optical Fiber, Radio

2. World Wide Web (WWW) and HTTP

HTTP Characteristics: Stateless client-server protocol over TCP (Default Port 80, HTTPS Port 443).

- **Non-Persistent HTTP (1.0):** Each object download requires a separate TCP connection (2 RTT per object + transmission time).
- **Persistent HTTP (1.1):** Reuses a single TCP connection for multiple web objects, drastically cutting connection establishment latency.
- **HTTP Methods:** GET (retrieve data), POST (submit form/payload), HEAD (retrieve headers only), PUT (upload/replace), DELETE.
- **HTTP Status Codes:** 200 (OK), 301 (Moved Permanently), 304 (Not Modified / Cached), 400 (Bad Request), 404 (Not Found), 500 (Internal Server Error).

- **Cookies & Web Caching:** Cookies provide state management over stateless HTTP via 4 components: Header in HTTP response, Header in next request, Cookie file on client host, Backend database.

3. File Transfer Protocol (FTP) & Electronic Mail

FTP Architecture: Uses two separate parallel TCP connections (Out-of-band control):

- **Control Connection (Port 21):** Carries user commands and server replies throughout the session. Remains open.
- **Data Connection (Port 20):** Opened dynamically for each file transfer and closed immediately after transfer finishes.

Electronic Mail Architecture:

- **SMTP (Simple Mail Transfer Protocol - Port 25/587):** Push protocol between client/server and server/server using direct TCP and 7-bit ASCII commands (HELO, MAIL FROM, RCPT TO, DATA, QUIT).
- **Mail Access Protocols (Pull):** POP3 (Post Office Protocol v3 - Port 110, downloads emails locally, stateless), IMAP4 (Internet Message Access Protocol v4 - Port 143, stateful, synchronizes folders across multiple devices).
- **MIME (Multipurpose Internet Mail Extensions):** Extends SMTP to carry multimedia non-ASCII content using Base64 encoding.

4. Domain Name System (DNS) & P2P / BitTorrent

DNS Architecture: Distributed hierarchical database running over UDP/TCP Port 53.

- **Server Hierarchy:** 13 Root DNS server clusters → Top-Level Domain (TLD) servers (.com, .org, .in) → Authoritative DNS servers.
- **Resolution Types:** *Recursive* (local resolver queries on behalf of client until resolved) vs *Iterative* (resolver receives referral addresses).
- **Resource Records (RR):** Type A (Host → IPv4), Type AAAA (Host → IPv6), Type NS (Domain → Authoritative Name Server), Type CNAME (Canonical Alias), Type MX (Domain → Mail Server).

Peer-to-Peer Paradigm & BitTorrent:

- **P2P Self-Scalability:** As users join, they bring both demand and upload capacity. BitTorrent divides files into 256 KB chunks.
- **Rarest First Policy:** Peers request chunks that are least copied among neighbors first to maximize piece variety.
- **Tit-for-Tat Choking Algorithm:** Peers unchoke top 4 uploaders offering highest rates. Optimistic unchoking tests random new peers every 30 seconds.

MODULE 2: Transport & Network Layers with Linux Kernel

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1. Transport Layer: UDP & TCP Protocols

Transport Layer Role: Logical process-to-process communication using 16-bit Port Numbers (0-65535).

- **UDP (User Datagram Protocol):** Connectionless, unreliable, 8-byte fixed header (Src Port, Dst Port, Length, Checksum). Uses 1's complement addition over pseudo-header + UDP header + payload.
- **TCP (Transmission Control Protocol):** Connection-oriented, reliable byte stream, full-duplex, flow-controlled, congestion-controlled.
- **TCP Header (20-60 Bytes):** Sequence Number, Acknowledgment Number, Header Length (Data Offset), Flags (URG, ACK, PSH, RST, SYN, FIN), Advertised Window (rwnd), Checksum, Urgent Pointer.
- **Connection Setup & Tear-down:** 3-Way Handshake (SYN → SYN-ACK → ACK) and 4-Way Handshake (FIN → ACK, FIN → ACK).

Key Concept / Exam Formula:

TCP Congestion Control Phases:

1. **Slow Start:** cwnd starts at 1 MSS, doubles every RTT (exponential growth: 1, 2, 4, 8...) until reaching ssthresh.
2. **Congestion Avoidance (AIMD):** When cwnd $\geq$ ssthresh, cwnd increases linearly by 1 MSS per RTT.
3. **On Loss Detection:**
 - *Timeout (Severe):* ssthresh = cwnd/2, cwnd = 1 MSS → enters Slow Start.
 - *3 Duplicate ACKs (Mild):* ssthresh = cwnd/2. In **TCP Reno**, cwnd = ssthresh + 3 MSS (Fast Recovery).

2. Socket Programming & I/O Multiplexing (Hands-on)

TCP Socket Lifecycle:

- **Server:** socket() → bind() → listen() → accept() → read()/write() → close()
- **Client:** socket() → connect() → write()/read() → close()

I/O Multiplexing (select vs poll):

- **select():** Monitors bit-masks of file descriptors (readfds, writefds) up to FD_SETSIZE (1024). Reconstructs fd_set on each loop.
- **poll():** Uses an array of struct pollfd (fd, events, revents). Eliminates the 1024 fd limit and avoids fd_set reconstruction.

3. Network Layer: IPv4, Subnetting & Unicast Routing

IPv4 Datagram Header (20-60 Bytes): Ver (4b), IHL (4b), Type of Service (8b), Total Length (16b), ID (16b), Flags [DF, MF] (3b), Fragment Offset (13b), TTL (8b), Protocol [TCP=6, UDP=17] (8b), Header Checksum (16b), Src IP (32b), Dst IP (32b).

Key Concept / Exam Formula:**Subnetting Calculations:**

Given prefix /n → Subnet mask has n leading 1s. Host bits = 32 - n.

- Number of usable hosts = $2^{(32 - n)} - 2$
- Network ID = IP bitwise AND Subnet Mask
- Directed Broadcast Address = IP with all host bits set to 1

Unicast Routing Protocols:

- **Distance Vector (RIP):** Based on Bellman-Ford algorithm. Metric = Hop count (Max 15 hops, 16 = infinity). Suffers from *Count-to-Infinity* (mitigated by Split Horizon & Poison Reverse).
- **Link State (OSPF):** Based on Dijkstra's Shortest Path First (SPF) algorithm. Floods Link State Advertisements (LSAs) within hierarchical Areas (Area 0 = Backbone). Metric = Cost (Bandwidth based).
- **Path Vector (BGP):** Inter-Autonomous System (Inter-AS) protocol. Exchanges complete AS-Path vectors to eliminate loops and enforce routing policies.

4. Multicast Routing, Next-Gen IP (IPv6) & QoS

- **Multicast Protocols:** IGMP (group joins/leaves). DVMRP (Reverse Path Forwarding with flood & prune), PIM-DM (dense mode flood/prune), PIM-SM (sparse mode Rendezvous Point shared tree).
- **IPv6 Enhancements:** 128-bit addresses (hex: 2001:db8::1), fixed 40-byte base header (no checksum, faster processing), built-in security (IPsec), SLAAC stateless auto-configuration.
- **QoS Traffic Shaping:** *Leaky Bucket* (enforces strict constant rate, discards excess bursts) vs *Token Bucket* (tokens accumulate periodically, allowing burst transmission up to token capacity).
- **Linux Routing Table Hands-on:** Kernel FIB (Forwarding Information Base). Command: `ip route add 192.168.10.0/24 via 10.0.0.1 dev eth0`, `ip route show`.

MODULE 3: Data Link Layer, Ethernet & Wireless LANs

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1. Data Link Control (DLC): Framing, Flow & Error Control

Framing Methods:

- **Character Count:** Specifies number of characters in header (single bit error desynchronizes all subsequent frames).
- **Byte Stuffing:** Uses FLAG byte (e.g. 0x7E) with ESC escape bytes when FLAG appears in data payload.
- **Bit Stuffing (HDLC):** Flag = 01111110. Sender inserts a '0' after every five consecutive '1's. Receiver strips the '0' following five '1's.

Key Concept / Exam Formula:

Cyclic Redundancy Check (CRC):

Given data bitstream M of length k and divisor polynomial $G(x)$ of degree r :

1. Append r zeros to M ($M * 2^r$).
2. Perform Modulo-2 binary division (using XOR operations).
3. The r -bit remainder R is the Frame Check Sequence (FCS).
4. Transmit M followed by R . Receiver divides by $G(x)$; if remainder is 0, frame is error-free.

2. Multiple Access Protocols (MAC)

- **Pure ALOHA:** Transmit immediately. Vulnerable period = $2 * T_{fr}$. Max throughput $S_{max} = 1/(2e) \approx 18.4\%$.
- **Slotted ALOHA:** Transmit only at slot boundaries. Vulnerable period = T_{fr} . Max throughput $S_{max} = 1/e \approx 36.8\%$.
- **CSMA/CD (Carrier Sense Multiple Access with Collision Detection):** Listen before and while transmitting. If collision occurs, transmit 32-bit jam signal and use **Binary Exponential Backoff** (wait $k * 512$ bit times, k in $[0, 2^n - 1]$).
- **Minimum Frame Size in CSMA/CD:** Frame transmission time must be at least $2 * \text{Propagation delay}$: $L_{min} = 2 * d_{prop} * \text{Bandwidth}$ (Standard Ethernet = 64 Bytes).

3. Link-Layer Addressing, Ethernet & Connecting Devices

- **MAC Address:** 48-bit (6 octets) globally unique physical address. First 24 bits = OUI (vendor), last 24 bits = NIC identifier.
- **ARP Protocol:** Broadcasts ARP Request (FF:FF:FF:FF:FF:FF) to find MAC of target IP. Destination replies with unicast ARP Reply.
- **Standard Ethernet Frame (IEEE 802.3):** Preamble (7B) + SFD (1B) + Dst MAC (6B) + Src MAC (6B) + Type (2B) + Payload (46-1500B) + CRC FCS (4B).
- **Learning Switches & VLANs:** Switches automatically build MAC tables by recording source MAC of incoming frames. VLANs (IEEE 802.1Q) partition a physical switch into multiple isolated broadcast domains.

4. Wireless LANs (IEEE 802.11) & Mobile IP

- **IEEE 802.11 Architecture:** Basic Service Set (BSS) → Extended Service Set (ESS) connected via Distribution System (DS).

- **CSMA/CA Protocol:** Uses DCF (Distributed Coordination Function) with SIFS < PIFS < DIFS inter-frame spaces and Network Allocation Vector (NAV).
- **Hidden & Exposed Node Problems:** Solved using optional RTS (Request to Send) / CTS (Clear to Send) 4-way handshake.
- **Mobile IP:** Mobile Node (MN) retains Home Address while assigning temporary Care-of Address (COA) via Foreign Agent (FA). Home Agent (HA) intercepts packets and tunnels them to FA (Triangle Routing).
- **Hands-on Raw Sockets:** `socket(AF_PACKET, SOCK_RAW, htons(ETH_P_ALL))` enables packet sniffing and frame injection at the link layer.

MODULE 4: Network Management & Physical Layer

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1. Network Management: SNMP & ASN.1

SNMP Architecture: Manager (NMS), Managed Agent, Management Information Base (MIB), and Structure of Management Information (SMI).

- **SNMP PDUs (UDP Port 161 Manager → Agent, Port 162 Agent → Manager):** GetRequest, GetNextRequest, GetBulkRequest, SetRequest, Response, Trap (unsolicited alarm), InformRequest.
- **SNMP Versions:** SNMPv1 (plaintext community string), SNMPv2c (GetBulkRequest), SNMPv3 (USM user security: SHA/MD5 authentication + AES encryption; VACM view access control).
- **ASN.1 (Abstract Syntax Notation One):** Formal language specifying data structures independently of machine architecture.
- **BER (Basic Encoding Rules):** Encodes ASN.1 using TLV (Type-Length-Value) triplet format.
- **MIB Tree & OID:** Hierarchical tree under iso.org.dod.internet.mgmt.mib-2 (1.3.6.1.2.1). E.g. sysDescr = 1.3.6.1.2.1.1.1.

2. Physical Layer: Data, Signals & Capacity Limits

Transmission Impairments: Attenuation (signal energy loss, measured in $\text{dB} = 10 \log_{10}(P_2/P_1)$), Distortion (phase shifts across harmonics), Noise (Thermal, Induced, Crosstalk, Impulse).

Key Concept / Exam Formula:

Theoretical Channel Capacity Formulas:

- **Nyquist Bit Rate (Noiseless Channel):** $C = 2 * B * \log_2(L)$ bps

Where B = Bandwidth in Hz, L = Number of discrete signal levels.

- **Shannon Capacity Theorem (Noisy Channel):** $C = B * \log_2(1 + \text{SNR})$ bps

Where SNR is linear power ratio: $\text{SNR}_{\text{dB}} = 10 * \log_{10}(\text{SNR}_{\text{linear}})$. E.g., 30 dB → SNR = 1000.

3. Digital Transmission: Line Coding Techniques

- **Unipolar NRZ:** 1 = High voltage, 0 = 0V. Simple, but large DC component and clock synchronization loss.
- **Polar NRZ-L vs NRZ-I:** NRZ-L (0 = Positive, 1 = Negative). NRZ-I (Transition at bit boundary for 1, no transition for 0).
- **Manchester Encoding (IEEE 802.3):** 0 = High-to-Low transition, 1 = Low-to-High transition in middle of bit interval. Self-synchronizing, requires 2x bandwidth.
- **Differential Manchester:** Transition always present at mid-bit for clocking; transition at beginning represents 0, no transition at beginning represents 1.
- **Bipolar AMI:** 0 = 0V, 1 = Alternating positive and negative voltages. Zero DC component; scrambled by B8ZS/HDB3 to prevent long runs of 0s.

4. Analog Transmission, Multiplexing & Media

- **Digital-to-Analog Modulation:** ASK (Amplitude Shift Keying), FSK (Frequency Shift Keying), PSK (Phase Shift Keying: BPSK, QPSK), QAM (Quadrature Amplitude Modulation: combines ASK + PSK; 16-QAM carries 4 bits/ baud, 64-QAM carries 6 bits/ baud).

- **Multiplexing:** FDM (Analog frequency bands), WDM (Optical light wavelengths over fiber), Synchronous TDM (Fixed time slots), Statistical TDM (On-demand time slots).
- **Guided Transmission Media:** Twisted Pair (UTP Cat 5e/6, STP), Coaxial Cable (copper conductor + mesh shield), Optical Fiber (Step-Index & Graded-Index, Single-mode vs Multi-mode; immune to EMI, immense bandwidth).
- **Unguided Media:** Radio Waves (Omnidirectional, wall penetrating), Microwaves (Line-of-sight parabolic dish), Infrared (Short-range line of sight).